CS-251-MJ: Object Oriented Programming using C++
Lectures: 30 (Credits-2)

Prerequisites: Knowledge of C Programming Language

Learning Objectives:

1. To learn principles of Object-Oriented Programming (OOP).
2. To learn operators, type casting, reference variable.
3. To understand concept of functions, default arguments.
4. To acquire the knowledge of function and operator overloading.
5. To learn concept of Inheritance, abstract class.

Course Outcomes: On completion of this course, students will be able to

CO1: Understand object-oriented concepts such as data abstraction, encapsulation, inheritance, dynamic binding and polymorphism.
CO2: Make use of this pointer and access specifiers.
CO3: Apply new and delete operator.
CO4: Understand concept of polymorphism.
CO5: Illustrate concept of inheritance, virtual function, pure virtual function.

Unit 1: Object oriented concepts **02 (CO1)**

1.1 Procedure-oriented programming Vs Object-oriented programming
1.2 Classes and objects
1.3 Abstraction
1.4 Inheritance
1.5 Polymorphism
1.6 Encapsulations

Unit 2: Introduction to C++ programming **07(CO2)**

2.1 Features of C++
2.2 Data Types and Operators
2.3 Type casting in C++
2.4 Reference variables, this pointer
2.5 Usage of namespace, Managing Console I/O, Usage of Manipulators
2.6 Access specifiers
2.7 Defining data members and member functions
2.8 Array of objects

Unit 3: Function in C++ **06(CO3)**

3.1 Constructor, types of constructors, Destructor
3.2 Memory allocation (new and delete)
3.3 Call by reference, Return by reference
3.4 Default arguments
3.5 Inline function
3.6 Static class members
3.7 Friend functions

Unit 4: Function overloading and Operator overloading **07 (CO4)**

4.1 Function overloading

4.2 Overloading unary operators and binary operators (with member function and with friend function)

4.3 Overloading insertion and extraction operator

Unit 5: Inheritance **08 (CO5)**

5.1 Inheritance and its types

5.2 Constructor and destructor in derived class

5.3 Virtual base classes and abstract base classes

5.4 Function overriding

5.5 Virtual functions and pure virtual function

Reference Books:

1. Object Oriented Programming (C++) Balaguruswamy, McGraw Hill Education; Seventh edition
2. The Complete Reference C++ by Herbert Schildt, McGraw Hill Education; 4 edition
3. Mastering C++ by Venugopal, T Ravishankar, McGraw Hill Education; 2 edition A Structured Programming Approach Using C, Behrouz A. Forouzan, Richard F. Gilberg, Cengage Learning India